

Dense and Convolutional Autoencoders for Real-Time Anomaly Detection and New Physics Searches at the LHC Level-1 Trigger

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Real-time anomaly detection at the Large Hadron Collider Level-1 Trigger is crucial for identifying new physics beyond the Standard Model. This report presents the results of a comparative study of dense and convolutional Autoencoders for unsupervised anomaly detection using particle-level data available at the 40 MHz collision rate. The project aims to explore the potential of model training using a custom, physics-informed masked loss function that incorporates transverse momentum p_T , pseudorapidity η , and azimuthal angle ϕ measurements, with weights that tune the reciprocal relevance of each observable. For every model, the performance is presented in terms of AUC and $\text{TPR@FPR}(10^{-5})$ and compared with other studies used as benchmarks.

I. INTRODUCTION

The High-Luminosity phase of the Large Hadron Collider (HL-LHC) aims to vastly expand the physics potential of its main experiments by increasing the delivered integrated luminosity. This major increment in statistics will enable precision measurements of Standard Model parameters and will improve the sensitivity to rare and Beyond Standard Model processes, increasing the potential to observe new phenomena at higher energy scales or lower cross sections.

With a collision rate of 40MHz, it's not feasible to store all the data coming from the collisions. Therefore, different levels of triggers are implemented to perform an efficient online selection on the events. The lower level, usually called Level-1 Trigger (L1T), operates at the collision rate and relies on selection rules optimized to detect Standard Model processes. However, these conventional approaches are inherently biased toward expected physics, potentially missing unexpected new physics signatures.

Efforts are being made to improve the selection by implementing compact, efficient deep learning models directly on the L1T FPGAs. Autoencoders are one of the most powerful tools to perform unsupervised anomaly detection. By learning from a large dataset of Standard Model (background) events, they can flag events whose reconstruction error exceeds typical levels, indicating deviation from the learned background topology. The advantage of this method is that it does not require labeled samples of exotic signals, making it powerful to detect unknown physics scenarios.

Autoencoders can be implemented with different architectures and different training scenarios. This report presents the results of a comparative study of dense and convolutional (variational) Autoencoders for anomaly detection at the trigger level using publicly available LHC datasets [1–5], employing a physics-informed loss function to try and optimize the training towards physically meaningful representations of the input data.

II. DATA

The datasets used in this project are publicly available and described accurately in Ref. [6]. They consist of a background dataset and four distinct signal datasets.

The background dataset consists of 13.451.915 events of Standard Model processes that represent typical proton-proton collisions: inclusive W boson production decaying to a charged lepton and a neutrino $W \rightarrow \ell\nu$, inclusive Z boson production decaying into two leptons $Z \rightarrow \ell\ell$, $t\bar{t}$ production and QCD multijet production. They have been pre-filtered requiring the presence of at least an electron or a muon with $p_T > 23$ GeV and $|\eta| < 3$ or $|\eta| < 2.1$ respectively for electrons and muons. These events are exclusively used for training and validation of the models according to an AE logic. The dataset is split as follows: 64% for training (8.6M events), 16% for validation (2.1M events) and 20% for testing (2.7M events).

The four signal datasets consist of different new physics scenarios of topologies and masses:

1. A neutral scalar boson with $m_A = 50$ GeV decaying to two off-shell Z bosons, each forced to decay into two leptons $A \rightarrow 4\ell$.
2. A charged scalar boson $h^\pm$ with $m_{h^\pm} = 60$ GeV decaying into a tau lepton and a neutrino $h^\pm \rightarrow \tau\nu$.
3. A neutral scalar boson h^0 with $m_{h^0} = 60$ GeV decaying to two tau leptons $h^0 \rightarrow \tau\tau$.
4. A leptoquark LQ with $m_{LQ} = 80$ GeV decaying into a beauty quark and a tau lepton $LQ \rightarrow b\tau$.

Spanning from Standard Model-like processes to exotic scenarios, they allow a comprehensive assessment of the models' anomaly detection ability.

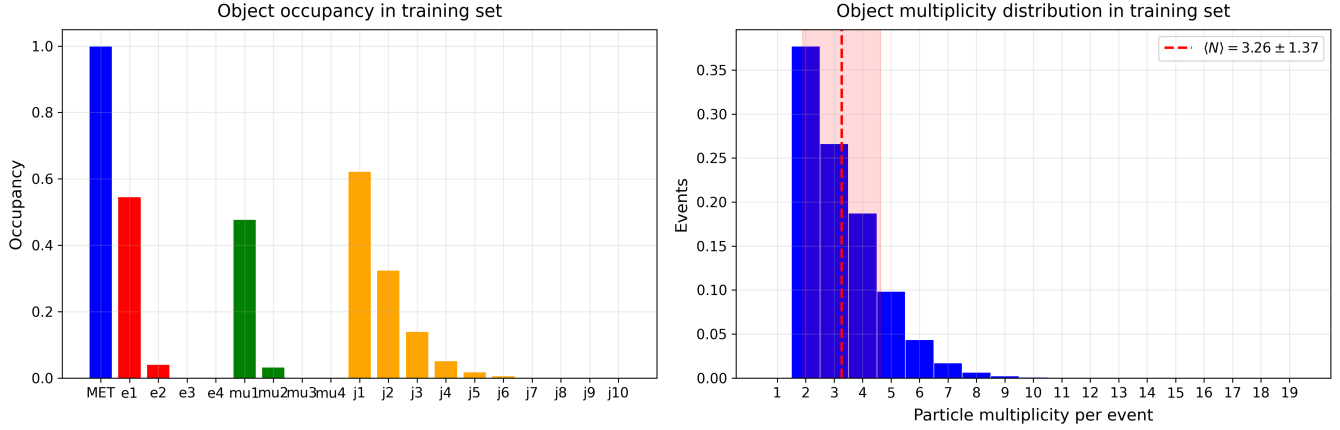


FIG. I. Particle occupancy distribution in background events. Nearly all events contain at most 10 actual particles, with a median occupancy of 3.26 particles per event.

A. Raw Datasets

The raw datasets available on Zenodo are compressed HDF5 files containing matrices with a $(N_{\text{events}}, N_{\text{particles}}, N_{\text{features}}) = (N_{\text{events}}, 19, 4)$ structure. For each event, to ensure consistency throughout the dataset, the number of particles is fixed at 19; for events containing fewer particles, slots are filled with zeros across all features (zero-padding). Each particle is characterized by four features: transverse momentum p_T [GeV], pseudorapidity η , azimuthal angle ϕ [rad] and particle class (0 for MET, 1 for electrons, 2 for muons and 3 for jets). The class distribution of the 19 particles in each event is fixed:

Particle index	Particle class
0	MET
1–4	e
5–8	μ
9–18	jets

This representation captures the essential information available to Level-1 Triggers while capable of working at 40 MHz.

B. Datasets Preparation

Raw datasets are processed into high-level datasets that can be directly used for model training and performance evaluation. Since this project studies dense and convolutional Autoencoders, two different transformations are performed:

- For the **dense** architecture, the input tensor of shape $(19, 3)$ is flattened into a 57-dimensional vector. The particle type label is excluded, while the kinematic features (p_T, η, ϕ) are retained for each particle.

- For the **convolutional** architecture, the event is kept in its original ordered structure and reshaped into a $(19, 3, 1)$ tensor, where the last dimension corresponds to a single input channel. As in the dense case, particle type information is removed. For 1D architectures the last dimension can be ignored, working with the original $(19, 3)$ tensor.

C. Datasets Exploration

Before proceeding with the Autoencoder models, a study of the datasets characteristics is performed to gain a more complete physical understanding of the variables involved.

Starting from the particle occupancy in the background dataset (Fig. I), it is clear that the majority of particle slots in each event are zero-padded. The mean occupancy is $\langle N \rangle = 3.26 \pm 1.37$ particles per event with nearly all events containing at most 10 particles. While electron and muon multiplicities are typically close to unity, jet multiplicity is significantly higher, consistent with the presence of QCD multijet production processes as described in Ref. [6].

Comparing the feature distribution densities (Fig. II), then, it is possible to observe topological differences between processes, mainly in the tails of MET and jets transverse momentum p_T . Differences in these distributions are the key to perform anomaly detection using Autoencoders. For instance, differences are less pronounced in the ϕ distribution rather than in p_T as described before.

Finally, it is possible to extract physics observables from these data: computing the dilepton invariant mass the Z peak emerges from SM background events, the invariant transverse mass of a lepton and MET shows the production of W bosons while the four-lepton invariant mass of the $A \rightarrow 4\ell$ dataset shows the presence of a scalar boson with $m_A = 50$ GeV.

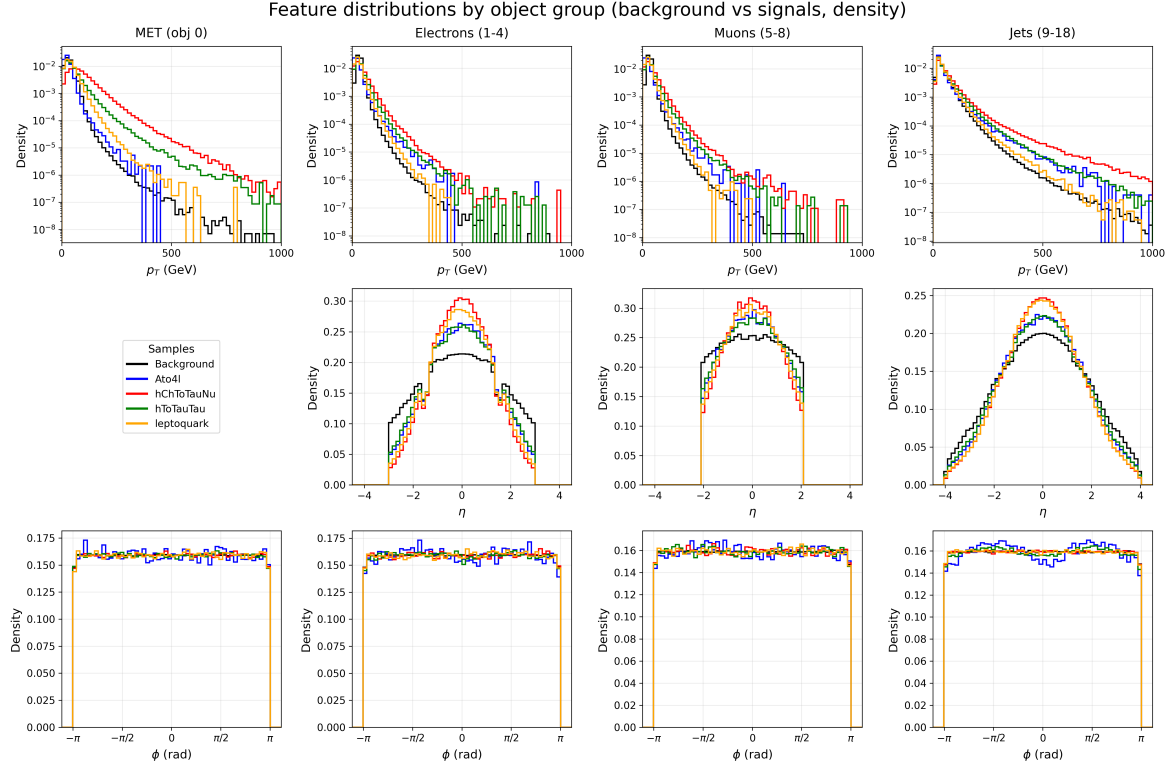


FIG. II. Normalized feature distribution comparing background and all signal channels. The density plots reveal topological differences between processes, mainly in the tails of MET and jets transverse momentum p_T .

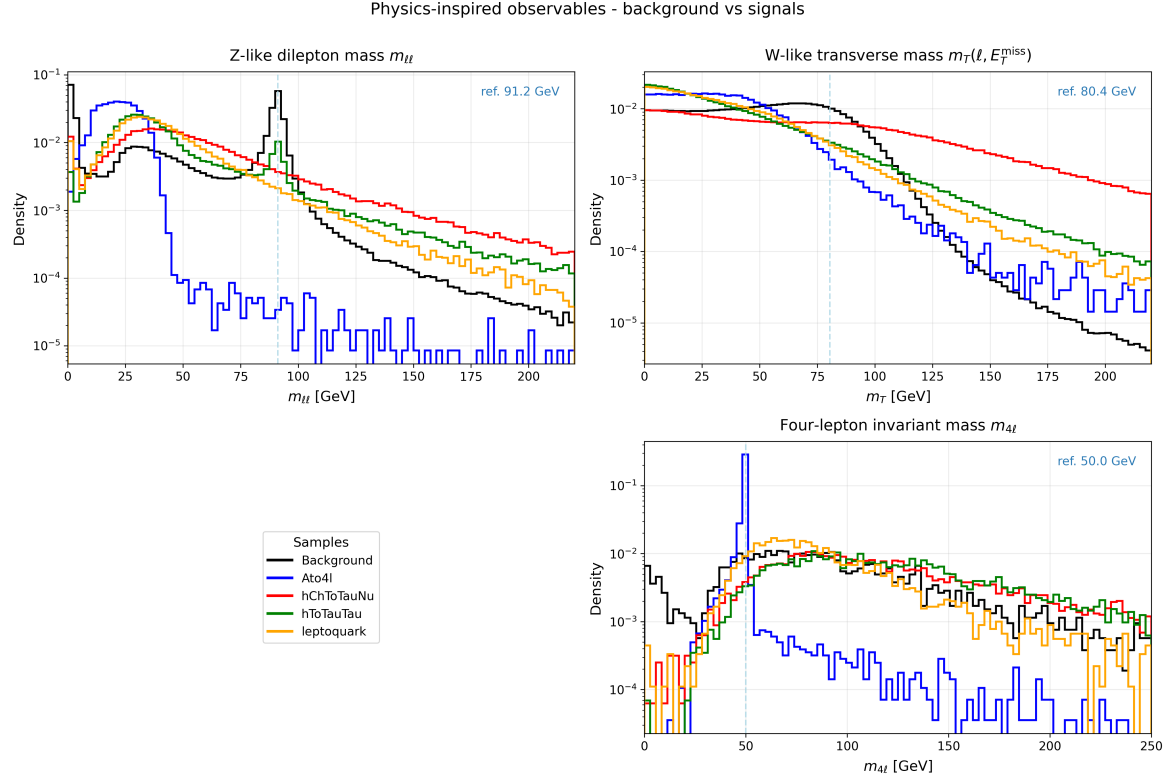


FIG. III. Distributions of reconstructed dilepton invariant mass, lepton MET invariant transverse mass and four-lepton invariant mass. It clearly emerges the production of Z and W bosons from background events and A (50GeV) from $A \rightarrow 4\ell$.

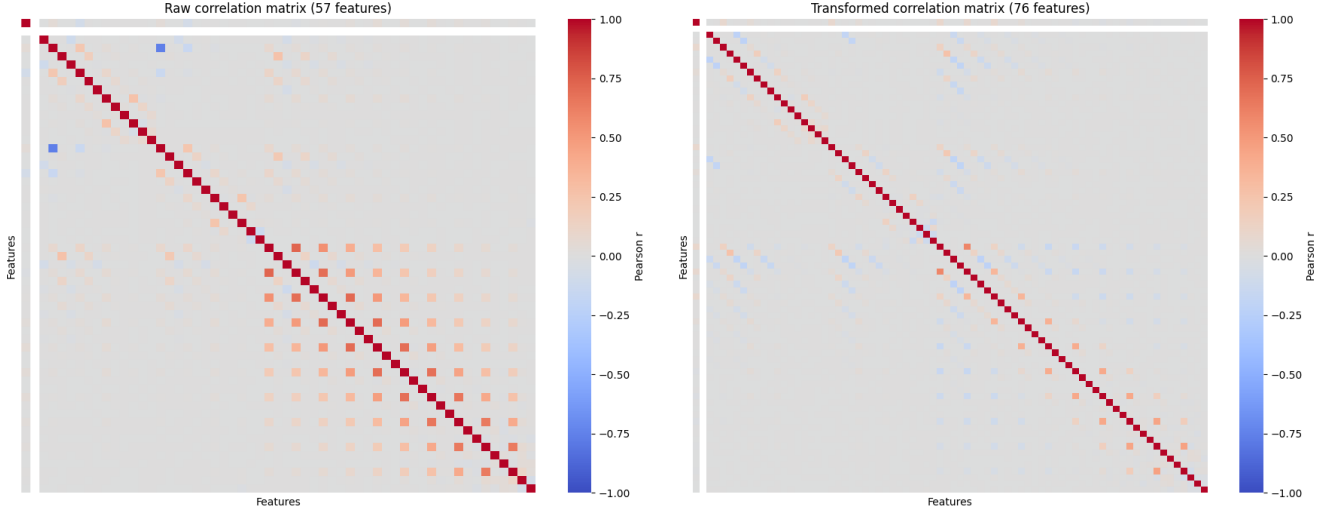


FIG. IV. Comparison between the correlation matrices of the raw dataset and the engineered one. The blank row and column correspond to the pseudorapidity of the MET, which is zero by definition.

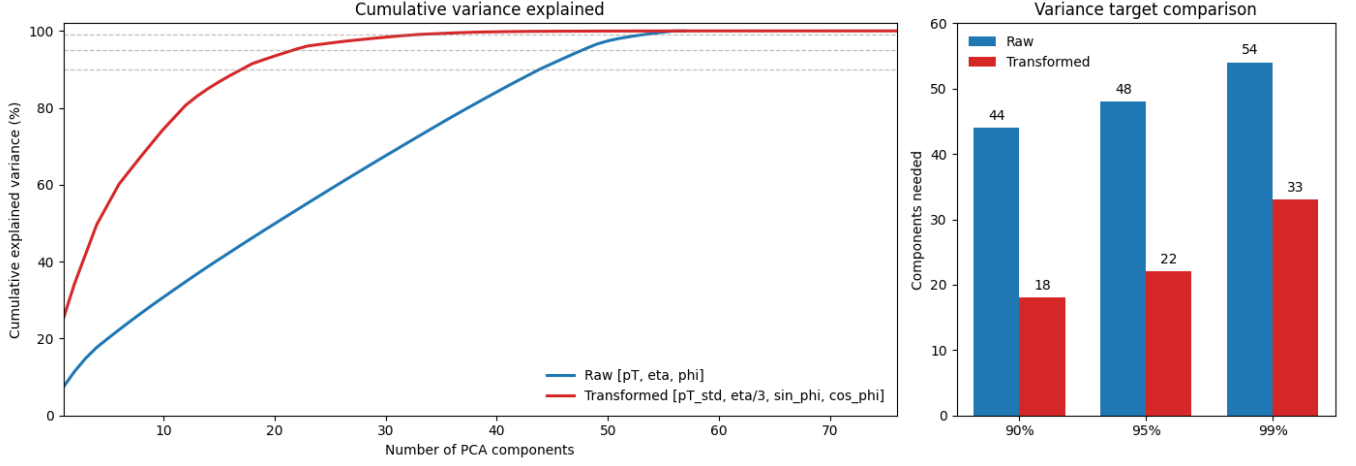


FIG. V. Cumulative variance comparison between the two datasets. It is clearly visible that feature engineering can lead to a better compression of the dataset into a lower-dimensional space

III. AUTOENCODERS

Since the model is designed for deployment at the L1T, introducing additional engineered features (e.g. invariant masses, jet-jet distances, or other high-level observables) or altering the existing feature set is not feasible due to strict latency and computational constraints. As a consequence, the model has limited access to explicit physics-informed representations of the events. A possible way to partially compensate for this limitation is to incorporate physics knowledge directly into the training procedure, for example through the use of a physics-oriented loss function rather than the standard MSE reconstruction loss, allowing the Autoencoder to prioritize physically meaningful event characteristics.

A. PCA

Principal Component Analysis (PCA) is used here as a diagnostic tool to gain insight into the potential benefits of incorporating physics-inspired transformations into the input representation. In particular, PCA allows one to quantify how efficiently the information contained in the dataset can be compressed into a lower-dimensional space, which is directly related to the performance expectations and latent space dimensions of Autoencoders and their variational versions.

The PCA decomposition of the original dataset is compared with that of a transformed dataset in which several physics-motivated preprocessing steps have been applied. These include the normalization of the transverse momentum p_T , the scaling of the pseudorapidity η by a factor of

1/3, and the replacement of the azimuthal angle ϕ with its sine and cosine representations, $\sin \phi$ and $\cos \phi$, in order to account for its periodic nature.

The results show that, although the original dataset already exhibits relatively weak correlations among variables, almost all principal components, considering the zero-padding, are required to explain 90% of the total variance. In contrast, the transformed dataset achieves the same level of explained variance with only 18 components. This indicates that the feature engineering procedure leads to a more compact and structured representation of the data.

This observation is particularly relevant in the context of Autoencoders, since their ability to compress information efficiently is strongly influenced by the intrinsic dimensionality of the input space. A representation that concentrates most of the variance in fewer effective directions is therefore expected to facilitate the learning of a more efficient latent space and improve the overall compression performance.

B. Physics-Informed Loss

Based on the considerations discussed above, the project investigates the impact of a custom physics-informed loss function on the performance of different models. The loss is defined at particle level and computed independently for each event. It is composed of three contributions corresponding to transverse momentum, pseudorapidity and azimuthal angle.

A binary mask is applied to exclude zero-padding particles (defined as particles with $p_T = 0$), ensuring that only physically meaningful objects contribute to the loss. The event-level normalization avoids biasing the training towards events with higher particle multiplicity. Finally, the total loss is averaged over all events in the batch.

1. Autoencoders loss

The Autoencoders are trained to reconstruct particle-level features using the following loss:

$$\mathcal{L} = \frac{1}{N_{\text{events}}} \sum_{a=1}^{N_{\text{events}}} \frac{\sum_{b=1}^{N_{\text{particles}}} m_{ab} \sum_c \omega_c \mathcal{L}_{ab}^c}{\max\left(\sum_{b=1}^{N_{\text{particles}}} m_{ab}, 1\right)}$$

with:

- $N_{\text{particles}} = 19$, corresponding to the fixed number of reconstructed particles per event;
- $c \in \{p_T, \eta, \phi\}$, representing the set of particle-level kinematic features;

- transverse momentum loss, normalized by its characteristic variance:

$$\mathcal{L}_{ab}^{p_T} = \frac{(p_{T_{ab}} - \hat{p}_{T_{ab}})^2}{\sigma_{p_T}^2}$$

- pseudorapidity loss, normalized by a fixed characteristic detector acceptance scale of 3:

$$\mathcal{L}_{ab}^{\eta} = \left(\frac{\eta_{ab} - \hat{\eta}_{ab}}{3} \right)^2$$

- azimuthal angle loss, defined using a periodic angular distance to account for the 2π periodicity:

$$\mathcal{L}_{ab}^{\phi} = \left[\text{atan2} \left(\sin(\phi_{ab} - \hat{\phi}_{ab}), \cos(\phi_{ab} - \hat{\phi}_{ab}) \right) \right]^2$$

- feature-dependent weights introduced to balance the relative importance of the reconstructed quantities:

$$\omega_c = (\omega_{p_T}, \omega_{\eta}, \omega_{\phi}) = (1.0, 1.0, 0.5)$$

- binary mask used to exclude zero-padding particles:

$$m_{ab} = \begin{cases} 1 & \text{if } p_{T_{ab}} \neq 0 \\ 0 & \text{if } p_{T_{ab}} = 0 \end{cases}$$

2. Variational Autoencoders loss

In the case of variational Autoencoders, the reconstruction loss is identical to the one defined above for standard Autoencoders. The main difference arises from the probabilistic formulation of the latent representation.

The encoder outputs the parameters of a latent Gaussian distribution, $\mu(x)$ and $\log \sigma^2(x)$, from which the latent variable is sampled using the reparameterization

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, 1)$$

The total loss is defined as

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{reco}} + \beta \mathcal{L}_{\text{KL}}$$

where β controls the strength of the latent regularization term.

The KL divergence between the approximate posterior $q(z|x)$ and the prior $p(z) = \mathcal{N}(0, 1)$ is given by:

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2} \sum_{j=1}^d (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2)$$

with d being the dimensionality of the latent space.

C. Models Architecture and Training

All the models studied in this work are implemented in TENSORFLOW using KERAS layers and Model.

The first and most substantial part of the work consists of a comparison among 12 different dense AE models trained on the background dataset by minimizing the loss function described in Sec. IIIB 1, using the Adam optimizer and the standard weights ω_c . Training is performed for up to 100 epochs with a batch size of 1024 and an initial learning rate of 10^{-3} , employing early stopping with a patience of 10 epochs and learning rate reduction on plateau, with the validation loss used as the monitored metric.

The best model is then trained with different weights $\omega_c = (1.0, 1.0, 1.0)$, $(1.0, 0.5, 0.5)$, $(0.5, 1.0, 1.0)$ as well as a standard MSE loss to better comprehend the impact of the physics-driven one.

This same model is then implemented as a dense VAE and trained in four different configurations of epochs for early stopping patience and KL weighting parameter (epochs, β) = $(10, 10^{-2})$, $(20, 10^{-2})$, $(10, 10^{-6})$ and $(20, 10^{-6})$.

This is followed by a brief investigations of convolutional architectures, including four different 2D AE models and one 1D model implemented both as an AE and a VAE. The training procedure follows the same structure adopted for the dense architectures.

IV. ANOMALY DETECTION PERFORMANCES

The table reporting all results is provided in the Appendix. As a reference benchmark, the models and results presented in Ref. [7] are considered, given that they adopt a similar training strategy based on a custom loss function.

A. Performance evaluation

The performance of the models is mainly evaluated by comparing the true positive rate at a fixed false positive rate threshold of 10^{-5} (TPR@FPR(10^{-5})) and the area under the ROC curve (AUC), using as anomaly score a distance function defined according to the same principles adopted in the physics-informed loss.

While the AUC provides a global measure of the model's discriminative power across all operating points, the TPR@FPR(10^{-5}) is more relevant in the context of Level-1 Trigger applications. In this regime, the operating point is constrained to extremely low false positive rates due to bandwidth and rate limitations. Therefore, the TPR@FPR(10^{-5}) directly quantifies the fraction of signal events that the model would be able to accept under realistic trigger conditions, making it a more meaningful figure of merit for comparing online selection performance.

Layer (type)	Output shape	Params
InputLayer (input)	(None, 57)	0
BatchNormalization	(None, 57)	228
Dense	(None, 48)	2,736
BatchNormalization	(None, 48)	192
LeakyReLU	(None, 48)	0
Dense	(None, 32)	1,536
BatchNormalization	(None, 32)	128
LeakyReLU	(None, 32)	0
Dense (latent)	(None, 16)	528
Dense	(None, 32)	512
BatchNormalization	(None, 32)	128
LeakyReLU	(None, 32)	0
Dense	(None, 48)	1,536
BatchNormalization	(None, 48)	192
LeakyReLU	(None, 48)	0
Dense	(None, 57)	2,793
Total params		10,509
Trainable params		10,075
Non-trainable params		434

TABLE I. Architecture of **dense_ae_10**.

B. Performance Comparison

1. Dense Autoencoders

Starting from the different dense models, three main considerations can be made:

1. As a benchmark, **dense_ae_1** adopts the same architecture as the dense Autoencoder presented in Ref. [7]. Despite the similar training strategy and loss formulation, its performance is significantly worse, highlighting a strong dependence of the results on the precise tuning of the model architecture and its interaction with the custom loss function.
2. The best-performing model overall is **dense_ae_10**, whose architecture and performance are illustrated in Tab. I and Fig. VI. This model achieves comparable results to the dense Autoencoder of Ref. [7] across all four channels. It features a relatively high-dimensional latent space and a smooth, gradually varying architecture ($57 \rightarrow 48 \rightarrow 32 \rightarrow 16 \rightarrow 32 \rightarrow 48 \rightarrow 57$), suggesting a controlled compression and reconstruction of the input space.

Particularly promising is the performance on the $A \rightarrow 4\ell$ channel, where the model reaches an AUC of 90% and a TPR@FPR(10^{-5}) of 3.6%. This channel is the most SM-like among the considered signals and features the cleanest experimental signature. The corresponding separation between signal and background is clearly visible in the distributions of the relevant physics observables shown in Fig. III.

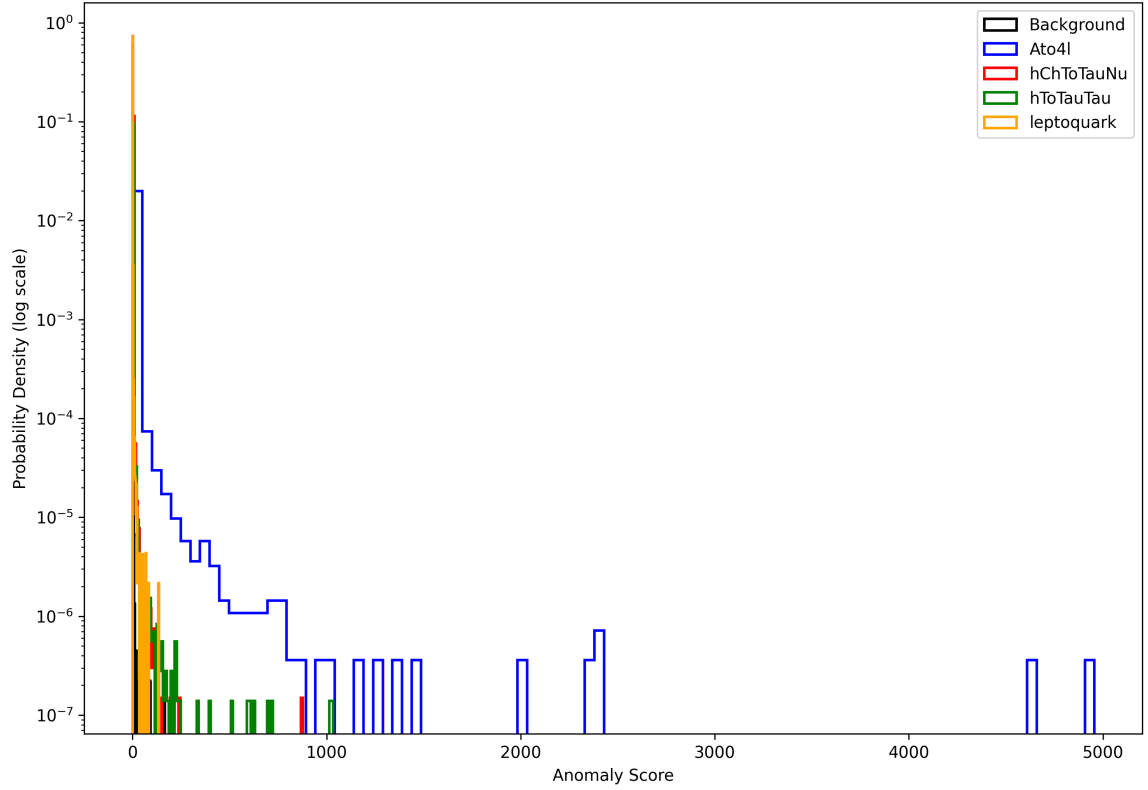
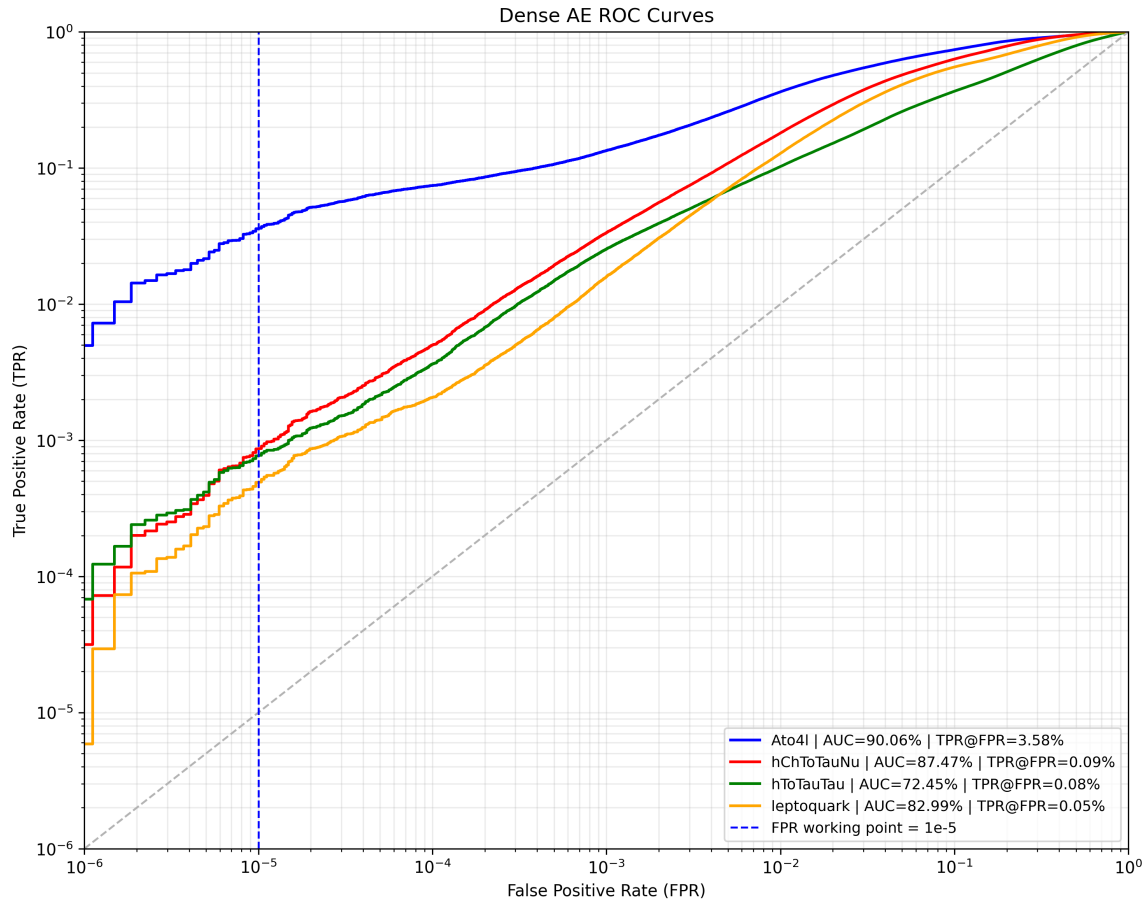


FIG. VI. ROC curve, TPR@FPR(1e-5) and Anomaly Score distribution for `dense_ae_10`.

The separation process done by the Autoencoder is also clearly visible looking at the anomaly score distribution between the background and the signals (Fig. VI).

This suggests that the model is particularly effective in regimes where the signal resembles well-understood Standard Model processes, benefiting from cleaner and more structured background distributions.

3. **dense_ae_5** and **dense_ae_6** introduce an asymmetric encoder-decoder structure and a narrower latent space ($57 \rightarrow 48 \rightarrow 32 \rightarrow 16 \rightarrow 8 \rightarrow 32 \rightarrow 57$ and $57 \rightarrow 32 \rightarrow 8 \rightarrow 16 \rightarrow 32 \rightarrow 48 \rightarrow 57$). Interestingly, they achieve the best performance on the $h^\pm \rightarrow \tau\nu$ dataset, even exceeding the results reported in Ref. [7]. Further studies are required to understand whether this improvement is due to a better latent representation or to an implicit regularization effect induced by the asymmetry.

The retraining of **dense_ae_10** with alternative weights and with a standard MSE loss function shows that increasing the relative importance of the p_T term improves the performance of the Autoencoder, at least in the discrimination of $A \rightarrow 4\ell$ signal from the background. In this channel, the $\text{TPR@FPR}(10^{-5})$ increases from 0.663% under MSE training to 3.606% when using the weighted loss with $(\omega_{p_T}, \omega_\eta, \omega_\phi) = (2.0, 0.5, 0.5)$.

This result provides clear evidence of the impact of a physics-informed loss function and highlights the importance of properly balancing the contributions of the different kinematic variables during training.

The variational version of this model, as described above, is trained under four different configurations of early stopping patience and regularization factor β . The overall performance is worse than the one of the corresponding Autoencoder. A lower β yields slightly better performance than a higher β in the $h^\pm \rightarrow \tau\nu$ channel, but no consistent improvement or effect is observed across the different training configurations.

2. Convolutional Autoencoders

Different configurations of convolutional Autoencoders are tested, both 1D and 2D. Convolutional neural networks are useful for processing information that contains spatial features and correlations. While p_T , η and ϕ can in principle be used to reconstruct the 3D kinematics of each event, this information is partially lost in the raw representation of the training dataset.

The performance of these convolutional models, while reasonably good in some cases in terms of AUC, is generally worse than that of the best dense model described above when considering the $\text{TPR@FPR}(10^{-5})$ metric. No significant difference in performance is observed between the 2D and 1D models.

The 1D model, also evaluated in its variational configuration, shows neither better nor worse performance compared to its standard version.

A possible future direction of this work would be the development of a loss function capable of better characterising spatial correlations between particles.

V. CONCLUSIONS

This project investigates the use of dense and convolutional Autoencoders for unsupervised anomaly detection at the LHC Level-1 Trigger, with particular focus on the impact of a physics-informed reconstruction loss. The study is performed on publicly available particle-level datasets including one Standard Model background sample and four Beyond Standard Model signal scenarios.

Model architecture and loss design both play a crucial role in low-FPR scenarios like trigger applications. The key outcome of the project is the observed benefit of physics-informed weighting in the loss function, demonstrating that embedding domain knowledge directly in the training process can materially improve trigger-oriented anomaly detection. Among the tested models, dense Autoencoders with high-dimensional latent spaces and smoothly varying architectures provided the most robust and competitive behavior across channels, with good results especially in the $A \rightarrow 4\ell$ channel where the clearest separation from background was achieved.

The variational extensions did not provide a consistent gain over standard Autoencoders in the explored configurations. Similarly, convolutional architectures achieved reasonable global discrimination in some cases but remained generally less effective than the best dense model in the $\text{TPR@FPR}(10^{-5})$ metric, suggesting that the current event representation may not fully exploit convolutional inductive biases.

Overall, these results support dense Autoencoders with physics-informed losses as a promising and practical direction for real-time anomaly detection at L1T.

VI. CODE AND DATA AVAILABILITY

All the code used in this project is written in `python>=3.11` using Jupyter Notebooks. The required packages are `h5py` `scikit-learn` `numpy` `matplotlib` `seaborn` `tensorflow` `keras==3.14.1`.

The data used in this project are openly available at Zenodo at Ref. [1–5].

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 - [2] T. Aarrestad, E. Govorkova, J. Ngadiuba, E. Puljak, M. Pierini, and K. A. Wozniak, Unsupervised New Physics detection at 40 MHz: $A \rightarrow 4\ell$ Signal Benchmark Dataset, 10.5281/zenodo.7152590 (2021).
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 - [7] E. Govorkova, E. Puljak, T. Aarrestad, T. James, V. Loncar, M. Pierini, A. A. Pol, N. Ghielmetti, M. Graczyk, S. Summers, J. Ngadiuba, T. Q. Nguyen, J. Duarte, and Z. Wu, Autoencoders on field-programmable gate arrays for real-time, unsupervised new physics detection at 40 MHz at the Large Hadron Collider, *Nature Machine Intelligence* **4**, 154–161 (2022).

Appendix: full model comparison of Autoencoders

Model	TPR @ FPR=1e-5 [%]				TPR @ FPR=1e-5 [Events]				AUC [%]			
	$A \rightarrow 4\ell$	$h^\pm \rightarrow \tau\nu$	$h^0 \rightarrow \tau\tau$	$LQ \rightarrow b\tau$	$A \rightarrow 4\ell$	$h^\pm \rightarrow \tau\nu$	$h^0 \rightarrow \tau\tau$	$LQ \rightarrow b\tau$	$A \rightarrow 4\ell$	$h^\pm \rightarrow \tau\nu$	$h^0 \rightarrow \tau\tau$	$LQ \rightarrow b\tau$
dense_ae.1	0.032	0.112	0.035	0.006	18	849	244	19	83.46	89.29	74.27	82.01
dense_ae.2	0.486	0.048	0.029	0.004	272	362	201	12	87.88	88.36	73.52	84.56
dense_ae.3	0.132	0.075	0.030	0.002	74	568	208	8	88.45	88.53	73.50	85.29
dense_ae.4	0.118	0.076	0.026	0.004	66	579	178	14	84.44	89.24	74.64	83.19
dense_ae.5	0.139	0.148	0.045	0.012	78	1125	313	40	87.68	88.95	74.06	83.48
dense_ae.6	0.100	0.186	0.081	0.011	56	1415	560	39	88.84	89.33	75.42	86.62
dense_ae.7	0.070	0.126	0.039	0.004	39	960	273	13	85.04	89.59	74.11	82.53
dense_ae.8	0.332	0.085	0.034	0.004	186	647	238	13	87.62	88.07	72.49	84.63
dense_ae.9	0.204	0.009	0.011	0.004	114	72	77	13	81.77	83.52	69.32	82.36
dense_ae.10	3.582	0.086	0.077	0.049	2005	657	531	167	90.06	87.47	72.45	82.99
dense_ae.11	1.093	0.115	0.043	0.012	612	875	297	42	87.87	88.39	74.85	85.30
dense_ae.12	0.177	0.066	0.027	0.008	99	505	188	28	90.11	87.55	70.90	81.71
dense_ae.10_loss0511	1.333	0.063	0.043	0.018	746	476	295	60	89.95	85.74	69.66	81.51
dense_ae.10_loss111	1.092	0.080	0.048	0.015	611	606	333	51	89.38	87.14	71.47	83.45
dense_ae.10_loss20505	3.606	0.062	0.057	0.032	2018	470	393	109	88.00	85.80	67.03	78.65
dense_ae.10_mse	0.663	0.029	0.038	0.007	371	220	260	25	88.18	84.22	71.07	82.48
dense_vae.0	0.490	0.069	0.035	0.018	274	524	239	60	89.19	90.28	74.95	85.88
dense_vae.1	0.189	0.083	0.030	0.008	106	630	206	27	87.81	90.31	74.97	85.97
dense_vae.2	0.390	0.096	0.031	0.003	218	730	213	11	79.58	88.13	68.01	74.51
dense_vae.3	0.127	0.095	0.027	0.004	71	725	186	14	81.57	87.08	68.69	76.52
conv2d_ae.0	0.197	0.076	0.019	0.005	110	581	128	16	88.18	87.95	74.02	85.28
conv2d_ae.1	0.470	0.046	0.056	0.011	263	353	387	37	90.49	89.05	73.34	82.81
conv2d_ae.2	0.005	0.010	0.005	0.000	3	74	35	0	53.22	62.52	53.53	50.37
conv2d_ae.3	0.432	0.033	0.044	0.011	242	251	305	39	91.84	89.51	75.22	84.98
conv2d_ae.4	0.007	0.039	0.015	0.001	4	299	103	4	78.80	86.80	74.26	81.90
conv2d_ae.4_mse	0.104	0.093	0.022	0.004	58	710	151	15	87.03	83.10	69.19	80.29
conv1d_ae.0	0.306	0.088	0.062	0.026	171	666	430	88	89.89	89.03	71.89	83.76
conv1d_vae.0	0.331	0.125	0.080	0.039	185	954	555	134	90.08	88.38	73.38	85.27